Elementary Analysis

— *EYNTKA* Series —

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sequences and series

HERE (Will look at other textbooks to see how this is implemented, but in many ways this is what analysis really starts with).

1.1 Sequences

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Real Numbers

In this chapter, I will cover the construction of the real numbers

2.1 Construction Of The Real Numbers

The natural numbers can be thought as essentially being defined on the human intuition of counting [1]. There are bigger sets of numbers, most of them generated by filling in gaps.

But what does it mean to fill in gaps? When you were first introduced to arithmetics, you were introduced to the basic operations: $+, -, \times, \div, \sqrt{}, \ln$.

However, Not all these operations will always return a value: in $\mathbb{N}$, there is no solution to the equation $3 - 5$, since $-2 \notin \mathbb{N}$ [2]. The integers extend the naturals making them closed under subtraction, that is

$$\forall a \in \mathbb{Z}\,, \exists b \in \mathbb{Z} \text{ such that } a + b = 0$$

Usually, b is denoted $-a$, so $a + (-a) = 0$. Conventionally, this is shorthanded to $a - a = 0$.

Similarly, $\mathbb{Q}$ can extends $\mathbb{Z}$ by making them closed under division, that is

$$\forall a \in \mathbb{Q}\,, \exists b \in \mathbb{Q} \text{ such that } ab = 1$$

Usually, b is denoted a^{-1} or $\frac{1}{a}$, so $aa^{-1} = 0$ or $\frac{a}{a} = 1$.

One might think this covers all possible "gaps" which existed before. However, we

[1]possible that other animals have the same intuition, like humming birds or horses, but right now, I'll stick to human's since as far as we know, humans are the only creatures that can conceptualize numbers beyond 20

[2]Or more precicely, no $a \in \mathbb{N}$ can be used as a solution to $3 - 5 = a$

Proposition 2.1.1: $\sqrt{2}$ is irrational

There does not exist a $a \in \mathbb{Q}$ such that $a = \sqrt{2}$

***Proof*:**
For the sake of contradiction, let's assume that $\sqrt{2}$ is rational. Let $a = \frac{p}{q}$ $(p, q \in \mathbb{Z})$ where $gcd(p, q) = 1$. If $gcd(p, q) \neq 1$, simply divide $\frac{p}{q}$ by $gcd(p, q)$. Since we're assuming $\sqrt{2}$ is rational:

$$\sqrt{2} = \frac{p}{q}$$

thus

$$2 = \frac{p^2}{q^2} \Rightarrow 2q^2 = p^2$$

From here, we'll take 2 cases:

p **is odd**: If p is even, then p^2 is odd. However, no matter what q is, $2q^2$ is even. Thus p cannot be odd

p **is even**: since $gcd(p, q) = 1$, q cannot be even, since then $gcd(p, q) = 2$. Thus, q is odd. Since p is even, it's divisble by 2. Thus p^2 is divisible by 4. However, $2q^2$ is only divisible by 2, meaning that p^2 and $2q^2$ must be different numbers, and hence p cannot be even.

However, all numbers in $\mathbb{Z}$ are either even or odd, thus we reached a contradiction.

Thus, $\mathbb{Q}$ is not closed under $\sqrt{}$. Furthermore, numbers like φ (the golden ration) is also not rational. We can fill in all these gaps by saying that any polynomial equation must have a solution. Thus

$$x^2 + 1 = 0 \Rightarrow x = \sqrt{2} \qquad x^2 - x - 1 = 0 \Rightarrow x = \varphi$$

However, this too has it's limits. In 1873, Euler proved that e cannot be the solution of a polynomial equation. Later, it was shown that $\pi, \ln(2)$, and e^{π} each cannot be the solution to a polynomial equation. So, a further class of numbers was invented: If we can devise an alrogirthm to determine a number, the number is *computable*. Essentially, all numbers you can think about are computable (e, π, etc.). However, there do actually exist *uncomputable* numbers, numbers that we know exist, but we cannot divise an alrogirthm to define it! An example of this would be the Chaitin number. So, we need to find a way of defining an even bigger cateogry of numbers, one with hopefully no gaps.

The way we'll do this is by defining a larger set, based off of $\mathbb{Q}$, which will represent all the gaps:

Definition 2.1.2: Dedikend Cut

A **cut** in $\mathbb{Q}$ is a pair of subsets A, B of $\mathbb{Q}$ st

1. $A \cup B = \mathbb{Q}$, $A \neq \emptyset$, $B \neq \emptyset$, $A \cap B = \emptyset$
2. if $a \in A$ and $b \in B$, then $a < b$
3. A contains no largest element

Let $x = A|B$ denote the [dedikend] cut

Example 2.1.3: here

1. $\{r \in \mathbb{Q} | r < 0\} \,|\, \{r \in \mathbb{Q} : r \geq 0\}$. Notice how this split kind of looks like $(-\infty, 0)|[0, \infty)$
2. $\left\{r \in \mathbb{Q} | r \leq 0 \text{ or } r^2 < 2\right\} \,|\, \left\{r \in \mathbb{Q} : r^2 \geq 2\right\}$. Notice how this split kind of looks like $(-\infty, \sqrt{2})|[\sqrt{2}, \infty)$
3. Note that $\pm\infty$ is not a Dedekind cut, since that would imply $\emptyset|\mathbb{Q}$, or $\mathbb{Q}|\emptyset$, but A and B cannot be empty.

Note that in most constructions, A is the more interesting set, and B contains the "rest" of the rationals.

We will give a name to this set of cuts:

Definition 2.1.4: Real Numbers

The set of all Dedekind cuts of $\mathbb{Q}$ is called the **real numbers**, denoted $\mathbb{R}$. A **real number** is a cut in $\mathbb{Q}$.

We will justify calling this set a set of numbers soon. We'll first show that this indeed contains the rationals, and that it no longer contains "gaps" (meaning it contains all previous numbers we talked about, and more).

Notice that every rational number can be constructed using a Dedekind cut:

2.1.5 or any $q \in \mathbb{Q}$, $A|B = \{r \in \mathbb{Q} | r < q\} \,|\, \{r in \mathbb{Q} | r \geq q\}$ is a Dedekind cut

Thus, we can identify every $q \in \mathbb{Q}$ with a $q^* \in \mathbb{R}$. In this way, we have that $\mathbb{Q}^* \subseteq \mathbb{R}$.

The next question is are there still gaps? Is there a place we can cut $\mathbb{R}$ to split it? This would mean we would need $A|B$, where $A \cup B = \mathbb{R}$, $A \cap B = \emptyset$, and $A, B \neq \emptyset$. The way we'll solve this is by first translating the order relation given to us in the definition of a dedikend cut to the real numbers:

Definition 2.1.6: defining an order relation on $\mathbb{R}$

If $x = A|B$ and $y = C|D$ are cuts such that $A \subseteq C$, then x is **less than or equal to** y and we write $y \leq y$. If $A \subseteq C$ and $A \neq C$, then x is **less than** Y and we write $x < y$.

It can easily be proven that this is indeed an order relation.

2.1.7 Note The definition is essentially only reliant on the first set in each dedikend cut. This further emphasizes that it's the main set we care about. The 2nd set can really be thought of as the "remainder" set

With the notion of an order, we can now also bound sets/numbers:

Definition 2.1.8: Upper And Lower Bound

An element $M \in \mathbb{R}$ is an **upper bound** for a set $S \subseteq \mathbb{R}$ (or S is bounded from above by M) if each $s \in S$ satisfies

$$s \leq M$$

Similarly for **lower bound**.

Definition 2.1.9: Least Upper Bound and Greatest Lower Bound

A *least upper bound* (or supremum) is an upper bound less than any other upper bound for S. We denote the least upper bound for S as $\sup(S)$.

Example 2.1.10: example 3

-1 is the least upper bound for the negative integers, while 4 is an upper bound.

Not all sets have a lest upper bound. For example, in $\mathbb{Q}$, there is no least upper bound for $\sqrt{2}$ (we will show this when we prove $\mathbb{Q}$ is dense in $\mathbb{R}$). A set that contains all it's least upper bounds is called a *complete* set. This is the key property that $\mathbb{R}$ has:

Theorem 2.1.11: $\mathbb{R}$ Is Complete

The set $\mathbb{R}$ is **complete** in the sense that it satisifes the least upper bound property: If S is a nonemtpy subset of $\mathbb{R}$ and is bounded above in $\mathbb{R}$, then there exists a least upper bound for S

***Proof*:**
Let $\mathcal{C} \subseteq \mathbb{R}$ be a nonempty subset of $\mathbb{R}$ that is bounded from above (so $\forall X|Y \in \mathcal{C}$, $X|Y < A|B$ for some $A|B \in \mathbb{R}$, or written in shorthand $\forall x \in \mathcal{C}, x < y$ for some $y \in \mathbb{R}$). Define the following dedikend cut:

$$\begin{aligned} C &= \left\{a \in \mathbb{Q} \middle| A|B \in \mathcal{C} \text{ such that } a \in A\right\} \\ D &= \text{ rest of } \mathbb{Q} \end{aligned}$$

Let $z = C|D$. It is easy to check that z is indeed a cut. We need to show that $z \leq y$ for any upper bound y. Indeed, let $y = C'|D'$ be any upper-bound for z. Then $C \subseteq C'$. Thus, $z \leq y$, showing that z is indeed the least upper bound

With this property, we can return to trying to make a dedikend cut of the real numbers. Let's say $\mathcal{A}|\mathcal{B}$ is a cut of the real numbers. By definition, $\mathcal{A}$ contains no largest elemetn. furthermore, each

$b \in \mathcal{B}$ is an upper-bound for $\mathcal{A}$. Therefore, there exists a $y = \sup(\mathcal{A})$. Therefore, every dedikend cut in $\mathbb{R}$ occurs at a real number!

2.1.1 Thinking of $\mathbb{R}$ as numbers

I mentioned earlier that $\mathbb{R}$ can be thought of as numbers. But what does it mean to be a number? The term "number" is actually an archatic term nowadays. Structures labeled as "numbers" are those which were historically labeled as numbers, since they were the ones studied the most. However, different types of numbers can have incredibly different properties. For example $\mathbb{N}$ and $\mathbb{R}$ are both labeled as numbers, but $\mathbb{N}$ and $\mathbb{R}$ are very different from one another! In mathematics, we have expanded this notion greatly in the domain of *Algebra*. For our purposes, we want $\mathbb{R}$ to have all the operation's we have learnt in arithmetics:

$$+ \qquad - \qquad \times \qquad \text{div}$$

and that they have all the properties we have grown accustom to:

1. $a + b = b + a$
2. $a + (b + c) = (a + b) + c$
3. $\exists 0 \in \mathbb{R}$ such that for every $a \in \mathbb{R}$, $0 + a = a$
4. $\forall a \in \mathbb{R}$, $\exists - a \in \mathbb{R}$ such that $a - a = 0$
5. $a \cdot b = b \cdot a$
6. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
7. $\exists 1 \in \mathbb{R}$ such that for every $a \in \mathbb{R}$, $1 \cdot a = a$
8. $\forall a \in \mathbb{R}$, $\exists a^{-1} \in \mathbb{R}$ such that $a \cdot a^{-1} = 1$
9. $a(b + c) = ab + ac$
10. if $x \leq y$ then $x + z \leq y + z$ for any $z \in \mathbb{R}$
11. [$\mathbb{R}$ is complete]

We have just shown the 10th and 11th property, so we'll go over some of the others. I say some, because it becomes quite tedious to prove all of them. They are all duable, but we will quickly start accumulating many very similar cases.

To define + such that $x + y = E|F \in \mathbb{R}$ works as we are familiar with, we define:

$$E = \{r \in \mathbb{Q} | \text{for some } a \in A \text{ and some } c \in C, \text{ we have } r = a + c\}$$
$$F = \text{ rest of } \mathbb{Q}$$

This operation is naturally well-defined. By definition, we also have that $x + y = y + x$.

With this same idea, we can define the zero cut $0*$, and show that $0 * +x = x$ for every $x \in \mathbb{R}$. We can also show associativity in the same way

To define $\times$, it becomes a bit harder, since there are cases when $x > 0*$ or $x < 0*$. Howver, besides that, it's the same principle.

Also how we say *the* real numbers because this construction is unique, making the word "number" a good use, since we usually think of numbers being the usual thing.

2.1.2 What about further extending $\mathbb{R}$?

comment on the hyperreals. They contain the reals. They do not form a metric space. they are an ordered topology though. Also no unique hyperreals

2.2 Cauchy and Convergent Sequences

We will take a second look at completeness. We have shown that number systems smaller than the real numbers leave some gaps. We tried showing those by what seemed like hand-wavy methods. I gave examples like golden ration, then π, then the chaitin number, to keep showing there are gaps, but that's an inconsistent way of finding gaps, especially for more general sets which we might want to prove there is not gap. Is there one way to test, one way to find gaps, that a system must satisfy in order to test to see whether a set has gaps? Furthermore, can we think of the concept of completeness beyond the idea of numbers? Are there more general spaces for which the concept of completeness applies? This is what we cover in this section.

The main idea that was used to capture this idea is that if we try to approach any potential "hole" with a sequence of numbers, that number *must* exist in our set. For example,

$$f : \mathbb{N} \to \mathbb{Q} \qquad x_n \in (\sqrt{2} - \frac{1}{n}, \sqrt{2} + \frac{1}{n}) \cap \mathbb{Q}$$

where x_n is some rational number inside the ever smaller open interval. Then this sequence approach $\sqrt{2}$, however $\sqrt{2} \notin \mathbb{Q}$, and it doesn't technically *converge* since for a sequence to converge, it must converge *to* a number. But in $\mathbb{Q}$, $\sqrt{2}$ is *not* a number, so it can't converge in it! We thus define what it means for a sequence to get arbitrarily close to "itself" as $n \to \infty$. [3].

Notice further that this new criterion for finding holes relies on sequences, in particular sequences that converge to one point. This might actually mean that completness is a notion that can be defined on spaces where there is a single point a sequence can approach. There are actually a couple more conditions that are also required; when taken together they define something called a uniform space. It's in the framework of uniform spaces that we can define something called *uniform properties*, which are properties that are *global* to the space. Completeness is a uniform property, since the entire space is uniform. If you heard of uniform continuity, this also is a uniform property (we will cover uniform continuity in section ref:HERE).

However, there is little advantage to build-up the formal construction of uniform spaces, since we would only use very briefly here to how completeness can be extended in full generality, but then not use it again (maybe forever in your mathematical career). Instead, we will prove how a subset of

[3] I will quickly point out that though I used a computable function to find the hole, and basically all examples one can think of will also find be computable, you can have uncomputable Cauchy functions, meaning numbers like the Chitin number are also considered

uniform spaces which we use much more often can be called complete. That subset is the class of metric spaces![4].

To start off with, let's formally define cauchy sequences in a metric space:

Definition 2.2.1: Cauchy Sequence

If (X, d) is a metric space, we say that a sequence (x_n) is a Cauchy sequence if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_n, x_m) < \epsilon$ whenever $n, m > N$.

Intuitively, this means that after we get far enough in the sequence, any two points will be arbitrarily close to each other. Cauchy sequences are more general than convergent sequences:

Example 2.2.2: ample 1

Take $X = \mathbb{Q}$, with the standard metric. Take the sequence:

$$x_1 = 1, x_{n+1} = 1 + \frac{1}{1 + x_n}.$$

Notice that $d(x_n, x_m) < \epsilon$, and therefor it's Cauchy, but $d(x_n, x) < \epsilon$ is not true because it's a infinite continued fraction, that will equal to the golden ratio φ. Since $\varphi \notin \mathbb{Q}$, it doesn't cenverge to any x. Another example taking advantage of different metrics is:

$$((0, 1], d_{Eu}),\ x_n = \tfrac{1}{n}$$
$$X = \mathbb{R}, d(x, y) = |e^x - e^y|, (x_n) = -n$$

However, all convergent sequences are cauchy:

Proposition 2.2.3: convergent sequence ⇒ Cauchy sequence

if (x_n) is a convergent sequence in (X, d), then (x_n) is Cauchy. (Note that being Cauchy does not mean it converges.)

Proof. Suppose $(x_n) \to x$ and fix an $\epsilon > 0$. Choose an $N \in \mathbb{N}$ such that $d(x_n, x) < \frac{\epsilon}{2}$ if $n > N$. This same N shows that (x_n) is Cauchy, for if $m, n > N$ then

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Which suffices to prove it. □

We can ask when the converse holds? That is, when is it guaranteed that when approaching a point, that point will exist? This essentially is the idea of completeness we formulated earlier said in a different way, since it says any time we approach what we might suspect is a gap, we can guarantee there is an element there! To formalize this intuition:

[4]Historically, completeness of metrics was studied first. Later, when the concept was abstracted, uniform spaces were defined in terms of pseudometrics, and were found to be the natural generalization for studying completeness, but they are rarely used for current lack of utility beyond their use in generalization

Theorem 2.2.4: $\mathbb{R}$ Is Complete Using Cauchy Sequences

$\mathbb{R}$ is *complete* with respect to Cauchy sequences in the sense that if (a_n) is a sequecne of real numbers which obeys a Cauchy condition then it converges to a limit in $\mathbb{R}$

***Proof*:**
P.18-19. Much shorter proof than in 257. Basically showing first that it's bounded ,than appropriately constructing a set that keeps bounded the sequence from above as $n \to \infty$, then take the sup of that set. Then prove that the cauchy sequence converges using the sup value.

This will become the de-facto way one thinks about completeness, since it generalizes the completeness proof presented in 2.1.11 for arbitrary metric spaces.

Notice though that it works for metric spaces, and that the proof relied on the metric at hand. You can actually then make cauchy sequecnes in spaces that seem to be complete, but are not (ex. look at the e^x metric on $[0, a]$ we defined earlier). You can even do more than that: you can have different metrics and construct different number systems! For exmaple, If we let p be a prime number, then let $d(x, y) = |x - y|_p$ where $|a|_p = \frac{1}{p^k}$ where p^k is the largest power of p that divides a, then this will define a different metric that is *not* the euclidean metric. This will in fact induce a different order on $\mathbb{Q}$ than the usual order, that is, it will no longer be that

$$1/2 \leq 1/5$$

but HERE. This links us back to our results from earlier in this chapter, were the construction of $\mathbb{R}$ was intimitely linked to the order of $\mathbb{Q}$.

Since it's a metric space, it will define cauchy sequences, and so we can compelte the space too! The completion of $\mathbb{Q}$ through this metric is called labeled $\mathbb{Q}_p$ and is called the p-adic numbers!

I remember hearing that the real numbers are the unique comlete ordered field, but the p-adics have the p-adic order, so I don't know how to combine the two.

2.3 Further properties of $\mathbb{R}$

In this section, we'll introduce a couple more proeprties the real numbers grant us.

Density

In topology, Given a space X, a set S is *dense* if $\overline{S} = X$. We can topologically prove that $\overline{\mathbb{Q}} = \mathbb{R}$. In real analysis, we would always be taking subsets $S \subseteq \mathbb{R}$ such that $\overline{S} = \mathbb{R}$. Density in $\mathbb{R}$ has a special property:

Proposition 2.3.1: Density In $\mathbb{R}$

A subset $S \subseteq \mathbb{R}$ is said to be dense in $\mathbb{R}$ if between any two real numbers, there exists an element of S. That is, if for any real number a, b such that $a < b$, then we have $S \cap (a, b) \neq \emptyset$

***Proof*:**
Exercice.

Example 2.3.2: density
The most famous such examples is that $\mathbb{Q}$ is dense in $\mathbb{R}$. A problem with doing.

Archemidean principle

Definition 2.3.3: Archimedean Property

Given an ordered space X, if for every $x \in X$, there is an integer n that is greater than x, then X has the *archemedean property*

Example 2.3.4: Archimedea
$\mathbb{R}$ has the archemedean property. For any $x \in \mathbb{R}$, there exists a $n = \lceil x \rceil + 1$

ϵ-principle

Finally, we'll label a fact that is very common, and you might've implictely internalized it alraedy doing continuity, convergent sequence, and limit proofs:

Theorem 2.3.5: ϵ Principle

If a, b are real numbers and if for each $\epsilon > 0$, we have $a \leq b + \epsilon$, then $a \leq b$. If for each $\epsilon > 0$, $|x - y| \leq \epsilon$, then $x = y$

***Proof*:**
easy to prove.

3

Differentiation

here

3.1 Analyticity and the Growth of Derivatives

Differentiability is a local, one-step condition, and smoothness iterates it without asking anything about how the successive derivatives behave. A power series expansion asks for much more: that the whole infinite list $f(a), f'(a), f''(a), \ldots$ determine f near a. This section measures the gap between the two, and finds that a single growth condition on the derivatives closes it.

Let f be smooth on an open interval. At a point a one may form the Taylor series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(t-a)^k$$

and ask whether it agrees with f on some interval around a.

Definition 3.1.1: Analytic Function

Let $U \subseteq \mathbb{R}$ be open. A function $f\colon U \to \mathbb{R}$ is *analytic* if for each $a \in U$ there is an $R > 0$ and real coefficients $c_0, c_1, c_2, \ldots$ such that

$$f(a+h) = \sum_{k=0}^{\infty} c_k h^k \qquad \text{for all } |h| < R$$

the series converging for every such h.

Every analytic function is smooth, since a power series may be differentiated term by term inside its interval of convergence, and that differentiation forces $c_k = f^{(k)}(a)/k!$: the only candidate expansion is the Taylor series. The converse fails, and the standard witness is

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Every derivative of this function at the origin vanishes, so its Taylor series there is

$$T_0 f = \sum_{k=0}^{\infty} 0 \cdot x^k = 0$$

which agrees with f at the single point $x = 0$ and nowhere else. The Taylor series exists, converges everywhere, and converges to the wrong function.

What goes wrong is visible in the remainder. Truncating the series after k terms splits f into a polynomial and an error,

$$f(t) = \underbrace{\sum_{j=0}^{k-1} \frac{f^{(j)}(a)}{j!}(t-a)^j}_{P_{k-1}(t)} + \underbrace{R_{k-1}(t)}_{\text{remainder}}$$

and the series converges to f at t precisely when $R_{k-1}(t) \to 0$. So the question is how large the remainder can be, and the following classical estimate answers it in terms of a single derivative.

Theorem 3.1.2: Lagrange Form of the Remainder

Let $k \geq 1$, let I be an open interval containing a, and let $f\colon I \to \mathbb{R}$ be k times differentiable. For each $t \in I$ with $t \neq a$ there is a point ξ_t strictly between a and t such that

$$f(t) = P_{k-1}(t) + \frac{f^{(k)}(\xi_t)}{k!}(t-a)^k$$

where P_{k-1} is the Taylor polynomial of degree $k-1$ for f at a.

***Proof*:**
Fix $t_0 \in I$ with $t_0 \neq a$. Since $(t_0 - a)^k \neq 0$, there is a unique constant Q with

$$f(t_0) = P_{k-1}(t_0) + \frac{Q}{k!}(t_0 - a)^k$$

and the claim is exactly that $Q = f^{(k)}(\xi)$ for some ξ between a and t_0. Define $F\colon I \to \mathbb{R}$ by

$$F(z) = f(t_0) - \sum_{j=0}^{k-1} \frac{f^{(j)}(z)}{j!}(t_0 - z)^j - \frac{Q}{k!}(t_0 - z)^k$$

At $z = t_0$ every term of the sum vanishes except the one with $j = 0$, which is $f(t_0)$, so $F(t_0) = 0$; and $F(a) = 0$ by the choice of Q. Each $f^{(j)}$ with $j \leq k-1$ is differentiable, so F is differentiable on I, and Rolle's theorem supplies a ξ_{t_0} strictly between a and t_0 with $F'(\xi_{t_0}) = 0$.

Computing that derivative, the product rule gives, for $1 \leq j \leq k-1$,

$$\frac{d}{dz}\left(\frac{f^{(j)}(z)}{j!}(t_0 - z)^j\right) = \frac{f^{(j+1)}(z)}{j!}(t_0 - z)^j - \frac{f^{(j)}(z)}{(j-1)!}(t_0 - z)^{j-1}$$

and the $j = 0$ term contributes $f'(z)$. Summing over j, each negative term cancels the positive term of the previous index, and only the last positive term survives:

$$\frac{d}{dz}\sum_{j=0}^{k-1}\frac{f^{(j)}(z)}{j!}(t_0 - z)^j = \frac{f^{(k)}(z)}{(k-1)!}(t_0 - z)^{k-1}$$

Differentiating the final term of F gives $Q(t_0 - z)^{k-1}/(k-1)!$, so

$$F'(z) = \left(Q - f^{(k)}(z)\right)\frac{(t_0 - z)^{k-1}}{(k-1)!}$$

Since ξ_{t_0} lies strictly between a and t_0, the factor $(t_0 - \xi_{t_0})^{k-1}$ is nonzero, so $F'(\xi_{t_0}) = 0$ forces $Q = f^{(k)}(\xi_{t_0})$, as we sought to show.

The remainder is therefore controlled by the size of the kth derivative on the interval, and nothing else. Fix a radius $r > 0$, write $I = [a - r, a + r]$, and put

$$M_k = \sup_{t \in I} |f^{(k)}(t)|$$

which is finite because $f^{(k)}$ is continuous on a compact interval. For $t \in I$ the Lagrange form bounds the remainder by

$$|R_{k-1}(t)| = \left|\frac{f^{(k)}(\xi_t)}{k!}(t - a)^k\right| \leq \frac{M_k}{k!}r^k$$

This is the worst the error can be, and whether it tends to zero is a question about the sequence $M_k/k!$ alone. The root test compares that sequence's decay against a geometric one, and its threshold quantity,

$$\limsup_{k\to\infty}\sqrt[k]{\frac{M_k}{k!}r^k} = r\limsup_{k\to\infty}\sqrt[k]{\frac{M_k}{k!}}$$

factors into the radius and an intrinsic property of f. That second factor is what the rest of the section turns on, so it gets a name.

Definition 3.1.3: Derivative Growth Rate

Let f be smooth on an open interval containing $I = [a-r, a+r]$, and set $M_k = \sup_{t\in I}|f^{(k)}(t)|$. The *derivative growth rate* of f on I is

$$\alpha_r = \limsup_{k\to\infty}\sqrt[k]{\frac{M_k}{k!}}$$

and f has *bounded derivative growth rate* on I if $\alpha_r < \infty$. It has *locally bounded derivative growth rate* on an open set U if every point of U lies in such an interval.

Example 3.1.4: Derivative Growth Rate of the Exponential

For $f = \exp$ on $I = [-r, r]$ every derivative is exp itself, so $M_k = e^r$ for all k and

$$\alpha_r = \limsup_{k\to\infty} \sqrt[k]{\frac{e^r}{k!}} = \lim_{k\to\infty} \frac{(e^r)^{1/k}}{\sqrt[k]{k!}} = 0$$

since $(e^r)^{1/k} \to 1$ while $\sqrt[k]{k!} \to \infty$. The growth rate is 0 at every radius, so $\alpha_r r = 0 < 1$ however large r is, and the Taylor series of exp converges to it on every interval. The same computation with $M_k = 1$ handles sin and cos.

With the notation in place the threshold quantity is $r\alpha_r$, and the root test converts a bound on it directly into convergence.

Theorem 3.1.5: Taylor Convergence Under Bounded Growth

Let f be smooth on an open interval containing $I = [a-\sigma, a+\sigma]$, with derivative growth rate α on I. If $\alpha\sigma < 1$ then the Taylor series of f at a converges to f uniformly on I, in the sense that

$$\sup_{|h|\le\sigma} \left| f(a+h) - \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} h^k \right| \longrightarrow 0 \qquad \text{as } n \to \infty$$

***Proof*:**

Choose $\delta > 0$ small enough that $(\alpha+\delta)\sigma < 1$. By the definition of the lim sup there is an N with

$$\sqrt[n]{\frac{M_n}{n!}} \le \alpha + \delta \qquad \text{for all } n \ge N$$

Let $|h| \le \sigma$. Applying theorem 3.1.2 at $t = a + h$ gives a θ between a and $a + h$ with

$$\left| f(a+h) - \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} h^k \right| = \left| \frac{f^{(n)}(\theta)}{n!} h^n \right| \le \frac{M_n}{n!}\sigma^n = \left(\sqrt[n]{\frac{M_n}{n!}}\, \sigma \right)^n$$

so for $n \ge N$ this is at most $\big((\alpha+\delta)\sigma\big)^n$. The bound does not depend on h, and $(\alpha+\delta)\sigma < 1$ makes it tend to 0, completing the proof.

The converse direction is the harder one: a function presented as a convergent power series must be shown to have derivatives that do not grow too fast. The obstacle is that differentiating a power series n times multiplies its coefficients by factors of size roughly k^n, and recovering a clean bound from that requires knowing how $n!$ compares with n^n. Two estimates carry the argument, and both are proved directly rather than quoted from Stirling's formula.

Theorem 3.1.6: Bounded Growth Rate of an Analytic Function

Suppose $f(a+h) = \sum_{k\ge0} c_k h^k$ for all $|h| < R$, and let $0 < \sigma < R$. Then f has bounded derivative growth rate on $I = [a-\sigma, a+\sigma]$.

***Proof*:**
Step 1: two factorial estimates. The first is

$$\lim_{n\to\infty} \frac{n}{\sqrt[n]{n!}} = e \tag{3.1}$$

Taking logarithms turns this into the claim that $\log n - \frac{1}{n}\log n!$ tends to 1. Applying the integral test to $\log n! = \sum_{j=1}^{n} \log j$,

$$\begin{aligned}
\frac{1}{n}\big(\log n^n - \log n!\big) &= \log n - \frac{1}{n}\big(\log n + \log(n-1) + \cdots + \log 1\big) \\
&\sim \log n - \frac{1}{n}\int_1^n \log x\, dx = \log n - \frac{1}{n}\Big(x\log x - x\Big)\Big|_1^n \\
&= 1 - \frac{1}{n}
\end{aligned}$$

which tends to 1, proving (3.1). The second is

$$0 < \lambda < 1 \quad \Rightarrow \quad \limsup_{n\to\infty} \sqrt[n]{\sum_{k=n}^{\infty} \binom{k}{n}\lambda^k} < \infty \tag{3.2}$$

Write $\lambda = e^{-\mu}$ with $\mu > 0$ and compare the sum with an integral:

$$\begin{aligned}
\sum_{k=n}^{\infty}\binom{k}{n}\lambda^k &= \sum_{k=n}^{\infty} \frac{k(k-1)\cdots(k-n+1)}{n!} e^{-k\mu} \\
&\le \frac{1}{n!}\sum_{k=n}^{\infty} k^n e^{-k\mu} \approx \frac{1}{n!}\int_n^{\infty} x^n e^{-\mu x}\, dx \\
&= \frac{-1}{n!} e^{-\mu x}\left(\frac{x^n}{\mu} + \frac{n x^{n-1}}{\mu^2} + \cdots + \frac{n!}{\mu^{n+1}}\right)\Big|_n^{\infty} \\
&\le \frac{1}{n!} e^{-\mu n}(n+1)n^n \frac{1}{\min(1,\mu)^{n+1}}
\end{aligned}$$

By (3.1) the nth root of $n^n/n!$ tends to e, while the nth roots of $e^{-\mu n}$, of $n+1$, and of $\min(1,\mu)^{-(n+1)}$ tend to $e^{-\mu}$, 1 and $1/\min(1,\mu)$ respectively. The whole nth root therefore tends to $e^{1-\mu}/\min(1,\mu)$, which is finite, proving (3.2).

Step 2: bounding the derivatives. Since $1/R$ is the lim sup of $\sqrt[k]{|c_k|}$, and $\sigma < R$, there is a λ with $\sigma/R < \lambda < 1$ and $|c_k|\sigma^k \le \lambda^k$ for all large k. Differentiating the series term by term n times, which is legitimate strictly inside the interval of convergence, gives for $|h| \le \sigma$

$$\begin{aligned}
|f^{(n)}(a+h)| &\le \sum_{k=n}^{\infty} k(k-1)\cdots(k-n+1)\,|c_k|\,|h|^{k-n} \\
&\le \frac{n!}{\sigma^n}\sum_{k=n}^{\infty}\binom{k}{n}|c_k|\sigma^k \le \frac{n!}{\sigma^n}\sum_{k=n}^{\infty}\binom{k}{n}\lambda^k
\end{aligned}$$

using $k(k-1)\cdots(k-n+1) = n!\binom{k}{n}$ and $|h|^{k-n} \le \sigma^k/\sigma^n$. Taking the supremum over $|h| \le \sigma$ and then nth roots,

$$\alpha = \limsup_{n\to\infty}\sqrt[n]{\frac{M_n}{n!}} \le \frac{1}{\sigma}\limsup_{n\to\infty}\sqrt[n]{\sum_{k=n}^{\infty}\binom{k}{n}\lambda^k}$$

which is finite by (3.2), as we sought to show.

The two directions now fit together, and the result is a characterization of analyticity that never mentions series at all.

Theorem 3.1.7: Analyticity Theorem

A smooth function on an open interval is analytic if and only if it has locally bounded derivative growth rate.

***Proof*:**
Suppose first that $f\colon (a,b) \to \mathbb{R}$ is smooth with locally bounded derivative growth rate, and let $x \in (a,b)$. Then x lies in an interval on which the growth rate α is finite, and shrinking that interval can only shrink the suprema M_k, hence cannot increase α. So choose $\sigma > 0$ small enough that $I = [x-\sigma, x+\sigma]$ sits inside it and $\alpha\sigma < 1$. By theorem 3.1.5 the Taylor series of f at x converges to f on I, so f is analytic at x.

Conversely, suppose f is analytic and let $x \in (a,b)$. Some power series $\sum c_k h^k$ converges to $f(x+h)$ for all h in an interval $(-R,R)$ with $R > 0$. Choosing any σ with $0 < \sigma < R$, theorem 3.1.6 shows f has bounded derivative growth rate on $[x-\sigma, x+\sigma]$, completing the proof.

The characterization makes the difference between the two classes quantitative. A smooth function is subject to no constraint at all on the size of its derivatives: given any sequence of reals whatsoever, a smooth function exists whose derivatives at a point realize it. An analytic function must instead satisfy a bound of the form
$$|f^{(k)}(x)| \leq C \cdot \frac{k!}{R^k}$$
on a neighbourhood, and it is exactly the failure of such a bound that lets e^{-1/x^2} be smooth while its Taylor series at the origin is useless.

4

Integration

here

4.1 Differentiation Under the Integral Sign

A fundamental question in analysis is identifying the conditions under which we can commute the integral and derivative operators. This is formalized by the Leibniz Integral Rule.

Theorem 4.1.1: Leibniz Integral Rule (Constant Limits)

Let $f(x,t)$ be a function defined for $x \in [a,b]$ and $t \in [c,d]$. Suppose that:

1. $f(x,t)$ is continuous on the rectangle $[a,b] \times [c,d]$.
2. The partial derivative $\frac{\partial f}{\partial t}$ exists and is continuous on $[a,b] \times [c,d]$.

Then, $F(t) = \int_a^b f(x,t)\,dx$ is differentiable on (c,d) and:

$$\frac{d}{dt}\left(\int_a^b f(x,t)\,dx\right) = \int_a^b \frac{\partial}{\partial t} f(x,t)\,dx$$

***Proof*:**
Let $F(t) = \int_a^b f(x,t)\,dx$. By the definition of the derivative:

$$F'(t) = \lim_{h\to 0} \frac{F(t+h) - F(t)}{h} = \lim_{h\to 0} \frac{\int_a^b f(x,t+h)\,dx - \int_a^b f(x,t)\,dx}{h}$$

By the linearity of the integral, we can combine the terms:

$$F'(t) = \lim_{h \to 0} \int_a^b \frac{f(x, t+h) - f(x,t)}{h} \, dx$$

We now fix x and consider $f(x,t)$ strictly as a function of t. By the Mean Value Theorem, for any fixed x, there exists a value $c_{x,h}$ strictly between t and $t+h$ such that:

$$\frac{f(x, t+h) - f(x,t)}{h} = \frac{\partial f}{\partial t}(x, c_{x,h})$$

We substitute this back into our expression. We wish to show that the limit of the integral approaches $\int \frac{\partial f}{\partial t}$. To do so, we analyze the difference:

$$\left| \int_a^b \frac{\partial f}{\partial t}(x, c_{x,h}) \, dx - \int_a^b \frac{\partial f}{\partial t}(x,t) \, dx \right|$$

Using the inequality $|\int g| \leq \int |g|$:

$$\leq \int_a^b \left| \frac{\partial f}{\partial t}(x, c_{x,h}) - \frac{\partial f}{\partial t}(x,t) \right| \, dx$$

Here we rely on the compactness of the domain. Since $\frac{\partial f}{\partial t}$ is continuous on the closed, bounded rectangle $[a,b] \times [c,d]$, it is *uniformly continuous.*

This implies that for any $\epsilon > 0$, there exists a $\delta > 0$ such that if the distance between two points is less than δ, the difference in function values is less than $\frac{\epsilon}{b-a}$.

Since $c_{x,h}$ is between t and $t+h$, we know $|c_{x,h} - t| < |h|$. If we choose $|h| < \delta$, then for all $x \in [a,b]$:

$$\left| \frac{\partial f}{\partial t}(x, c_{x,h}) - \frac{\partial f}{\partial t}(x,t) \right| < \frac{\epsilon}{b-a}$$

Substituting this bound back into the integral:

$$\int_a^b \left| \frac{\partial f}{\partial t}(x, c_{x,h}) - \frac{\partial f}{\partial t}(x,t) \right| \, dx < \int_a^b \frac{\epsilon}{b-a} \, dx = (b-a) \cdot \frac{\epsilon}{b-a} = \epsilon$$

Thus, as $h \to 0$, the difference goes to 0, proving the equality.

4.1.2 Note If the limits of integration also depend on t, i.e., $a(t)$ and $b(t)$, and are differentiable, the general Leibniz rule applies:

$$\frac{d}{dt} \int_{a(t)}^{b(t)} f(x,t) \, dx = f(b(t), t) \cdot b'(t) - f(a(t), t) \cdot a'(t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t} \, dx$$

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